VANKAYALA JAGATH CHARAN

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Machine Learning Engineer | Computer Vision & Edge AI | Generative AI (RAG) | Python Backend | MLOps

PROFESSIONAL SUMMARY

Machine Learning Engineer with 17+ months of startup experience building and deploying production AI products. Skilled in Computer Vision, Edge AI, Generative AI (RAG and LLM applications), Python and FastAPI backend development, React front ends, MLOps (Docker, CI/CD, MLflow, DVC), real-time video streaming, and data engineering and analytics (Spark, Snowflake, Power BI). Owns the full lifecycle from requirements to deployment, monitoring, and customer support across retail, industrial safety, and residential security.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Bash, TypeScript, Data Structures & Algorithms

ML / Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, Transformers, GANs, ONNX, TensorRT, Model Training & Evaluation, Model Optimization (FP16 Quantization), Dataset Curation

Computer Vision & Edge AI: OpenCV, MediaPipe, YOLO (v4/v5/v8), SSD, Object Detection, Face Recognition, Pose Estimation, Image Segmentation, Real-Time Video Analytics, NVIDIA Jetson, Raspberry Pi, CUDA, cuDNN, Embedded AI, IoT Integration

Generative AI & NLP: RAG, RAG Evaluation, LLMs (OpenAI, Hugging Face, LLaMA), LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, Vector Databases (PostgreSQL pgvector, Redis, MongoDB Atlas Vector Search), Embeddings, Chunking, Semantic Search, Hybrid Search, Semantic Ranking, Prompt Engineering, Conversational AI, NLTK

Backend & Frontend: FastAPI, Django, Flask, Asynchronous Programming (asyncio), REST API Development, Microservices, GraphQL, Pytest (Unit Testing), Swagger/OpenAPI, JWT/OAuth, API & System Integration, Twilio, React, Next.js, Streamlit

Data Engineering: Apache Spark (PySpark), Apache Kafka, Snowflake, Databricks, AWS Glue, ETL Pipelines, Data Modeling, Apache Airflow, AWS S3, MySQL, PostgreSQL, Label Studio, LabelImg

Data Analytics & BI: Power BI, Tableau, Excel, Data Visualization, Statistics, Exploratory Data Analysis (EDA), Dashboards & Reporting

Video Streaming: FFmpeg, H.264/H.265, RTSP, WebRTC, FLV, HTTPS Streaming, Low-Latency Pipelines

MLOps, DevOps & Cloud: Docker, Kubernetes, Terraform, Kubeflow, CI/CD (GitHub Actions, Jenkins, GitLab CI), MLflow, DVC, Model Deployment, Model Versioning, Monitoring (Prometheus, Grafana), OTA Updates, AWS (EC2, Lambda, S3), Google Cloud (Vertex AI, Gemini API), Cloudflare, Linux (Ubuntu), Git

Architecture & Collaboration: Modular AI Pipeline Design, Edge Server Deployments, Encrypted Deployment, Agile/Scrum, Technical Interviewing, Mentoring, Client Training

PROFESSIONAL EXPERIENCE

AMVAR DATA TECH PVT. LTD. (Startup) | **Software Developer – Machine Learning** | Oct 2024 – Present

- **Edge AI Server Platform:** Architected and launched a flagship platform with real-time object detection (YOLO), pose estimation, facial recognition, and audio event detection, serving 50+ enterprise clients at under 200 ms latency on edge hardware.
- Designed modular AI pipelines for retail (customer analytics, theft prevention), industrial safety (PPE detection, hazard monitoring), and residential security (intrusion detection, visitor management); deployed on NVIDIA Jetson and Raspberry Pi with sensor integration and robotic arm control.
- **Facial Recognition:** Developed an offline CNN face recognition system with 98.5% accuracy on custom datasets from video footage; TensorRT FP16 quantization delivered 30 FPS on Jetson Nano. Built annotation tools and automated dataset-generation workflows, cutting data preparation time by 60%.
- **Generative AI:** Built a production Retrieval-Augmented Generation (RAG) application using LLMs, embeddings, vector databases, and semantic ranking for enterprise document search, with offline operation, multi-language support, and domain adaptation for legal and technical documents.
- Built an emotionally aware voice assistant with FastAPI and Twilio, using a custom memory model for context retention and multilingual real-time interaction.
- **Video Streaming:** Engineered a low-latency FFmpeg (H.264/H.265) pipeline supporting RTSP, WebRTC, FLV, and HTTPS at under 200 ms end-to-end latency with GPU-accelerated live object detection; built secure remote access for on-premise servers without public IP exposure or cloud relay.
- Optimized inference with CUDA, cuDNN, TensorRT, and ONNX Runtime for a 35% latency reduction and 40% throughput gain.

- **Backend & Data:** Developed Python and FastAPI REST APIs and React applications integrating AI models, devices, and video sources into customer-ready products. Designed an ETL pipeline from AWS S3 to MySQL with Airflow scheduling and data profiling, and built Power BI dashboards for market share, sales calls, and territory performance.
- **MLOps & Security:** Built a secure deployment framework with end-to-end encryption, tamper protection, and health monitoring across 200+ edge devices, plus an OTA update system with zero-downtime rollouts; protected source code and model weights against reverse engineering.
- Built CI/CD pipelines (GitHub Actions, Docker) for automated testing, model versioning, and deployment validation; established MLflow and DVC workflows that improved model iteration velocity by 50%.
- **Customer & Team:** Represented the company at India Electronics Expo and IndiaSoft 2026, delivering technical demonstrations to 500+ stakeholders and conducting 20+ sales demonstrations and client training sessions.
- Conducted 15+ technical interviews, helped establish code review and training practices for a team of 10+ developers, and delivered Agile sprints 15% ahead of schedule.

EDUCATION

B.Tech, Computer Science and Engineering | M S Ramaiah University of Applied Sciences, Bengaluru | 2021 – 2025. Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures, Algorithms, Database Systems

CERTIFICATIONS

- Google Gen AI Academy and Google Cloud Skill Badges: Vertex AI, Gemini API, Imagen, Streamlit, Multimodal RAG
- End-to-End MLOps Bootcamp (Krish Naik, Udemy): MLflow, DVC, Docker, Airflow, SageMaker, Hugging Face
- Machine Learning Specialization (Andrew Ng, Coursera); Advanced Python Development (Udemy)

ADDITIONAL INFORMATION

Languages: English (Professional), Telugu (Native), Hindi (Fluent), Kannada (Fluent), Tamil (Conversational) | **Interests:** Agentic AI Systems & Workflow Automation, Edge AI Optimization